



# ABHAY MAHAJAN

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# Systematic Counting





2 Methods

$$\frac{4!}{3!} = 4$$

Different Questions

**Q1.** The number 3 can be written as a sum of positive integers in 4 ways viz. 3, 2 + 1, 1 + 2, 1 + 1 + 1. In how many ways can the number 5 be expressed?

$$\begin{array}{c} 1 + 1 + 1 \\ 1 \quad 1 + 1 \\ 1 + 1 \quad 1 \end{array}$$

$$\begin{array}{r} 311 \\ 131 \\ 113 \end{array} \left[ \begin{array}{l} 2+1 \\ 1+2 \\ 1+1+1 \end{array} \right]$$

$$(2!)$$

$$\frac{3!}{2!}$$

5

M1: Systematic Counting

$$5 - (1)$$

$$41 - (2)$$

$$32 - (2)$$

$$311 - (3)$$

$$\begin{array}{r} 221 - (3) \\ 2111 - (4) \\ 11111 - (1) \end{array}$$

$$\underline{\quad\quad\quad} (16)$$

M2: DP

$$\begin{array}{c} (1) + (1) + (1) + (1) \\ 1 + 3 + 1 \quad (16) \\ 1 + 1 \quad 1 + 1 + 1 \end{array}$$

**Q1.**The number 3 can be written as a sum of positive integers in 4 ways viz.  $3$  ,  $2 + 1$  ,  $1 + 2$  ,  $1 + 1 + 1$ . In how many ways can the number 5 be expressed?

**Answer: 16**

### Solution:

|     |                   |   |   |
|-----|-------------------|---|---|
| 5 = | 5                 | → | 1 |
|     | 1 + 4             | → | 2 |
|     | 2 + 3             | → | 2 |
|     | 1 + 1 + 3         | → | 3 |
|     | 1 + 2 + 2         | → | 3 |
|     | 1 + 1 + 1 + 2     | → | 4 |
|     | 1 + 1 + 1 + 1 + 1 | → | 1 |

---

Total = 16

$$8 \cdot 9 \left( \frac{17}{6} + 1 \right) = \frac{8 \cdot 9 \cdot 23}{6} = 23 \times 12$$

Systematic Counting

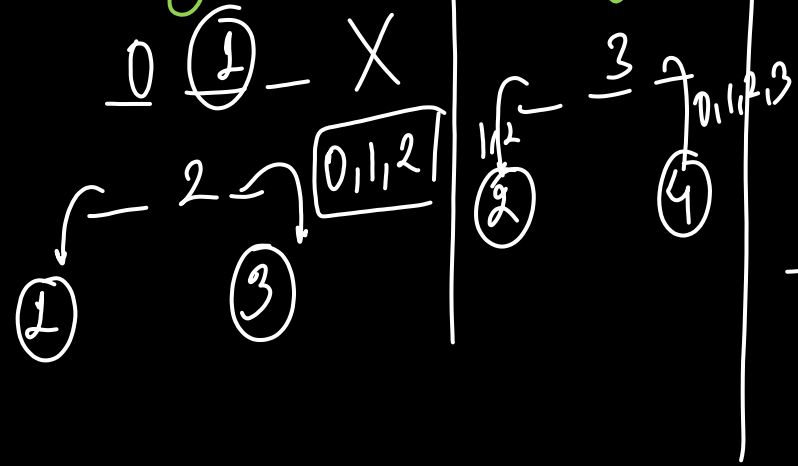
Q2. The number of three digit numbers of the form xyz such that  $x < y$  and  $z \leq y$  is N then  $(N - 225)$  is equal

$$276 - 225 = 51$$

$$276$$

$$x < y \geq z$$

M1 - Systematic Counting

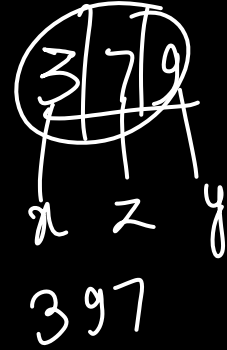


$$\begin{aligned}
 - 2 &\rightarrow 1 \cdot 3 \\
 - 3 &\rightarrow 2 \cdot 4 \\
 - 4 &\rightarrow 3 \cdot 5 \\
 &\vdots \\
 - 9 &\rightarrow 8 \cdot 10 \\
 \hline
 &8 \cdot 9 \cdot 17 + 8 \cdot 9
 \end{aligned}$$

$$\begin{aligned}
 &1 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 + \dots + 8 \cdot 10 \\
 &= \sum_{n=1}^8 n(n+1) \\
 &= \sum_{n=1}^8 (n^2 + n) \\
 &= \frac{2(8+1)(8+1)}{6} + \frac{2(8+1)(8+1)}{2}
 \end{aligned}$$

① 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

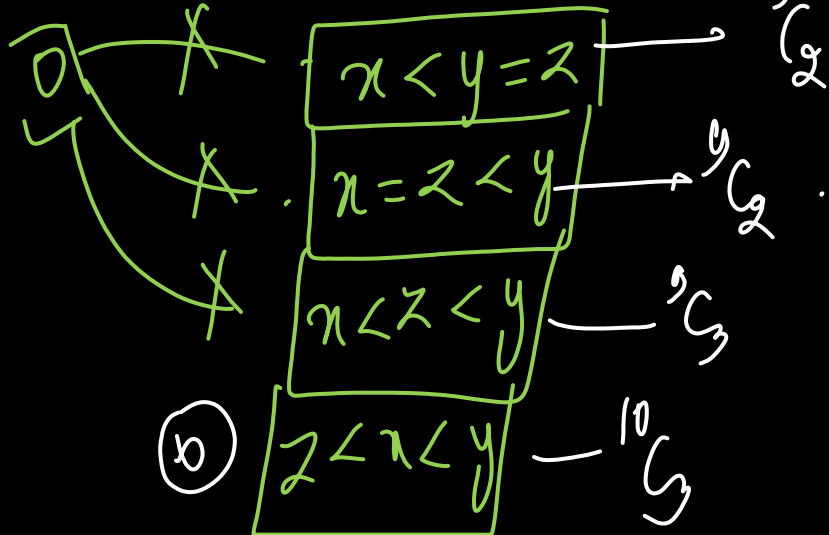
Q2. The number of three digit numbers of the form  $xyz$  such that  $x < y$  and  $z \leq y$  is  $N$  then  $(N - 225)$  is equal



Answer: 51

M2

$$x < y \geq z$$



$$2 \cdot {}^9C_2 + {}^9C_3 + {}^{10}C_3$$

276



## Solution:

$x < y$  and  $z \leq y$

|  |   |  |
|--|---|--|
|  | 4 |  |
|--|---|--|

 = 15,  

↓

3

↓

5

|  |   |  |
|--|---|--|
|  | 8 |  |
|--|---|--|

 = 63,  

↓

7

↓

9

|   |   |   |
|---|---|---|
| x | y | z |
| 1 | 2 |   |

 = 3,  

↓

3

|  |   |  |
|--|---|--|
|  | 5 |  |
|--|---|--|

 = 24,  

↓

4

↓

6

|  |   |  |
|--|---|--|
|  | 9 |  |
|--|---|--|

 = 80,  

↓

8

↓

10

|  |   |  |
|--|---|--|
|  | 3 |  |
|--|---|--|

 = 8  

↓

2

↓

4

|  |   |  |
|--|---|--|
|  | 6 |  |
|--|---|--|

 = 35,  

↓

5

↓

7

Total = 276

|  |   |  |
|--|---|--|
|  | 7 |  |
|--|---|--|

 = 48  

↓

6

↓

8

$$b+c=4$$

$$(1, 2, 0), 2, 0, 1, 2, 1, 0$$

$$b+c=2$$

$$\begin{array}{r} 102 \\ 201 \\ 210 \end{array} \quad \begin{array}{r} 102 \\ 201 \\ 210 \end{array}$$

**Q3.** Find the number of three -digit numbers from 100 to 999 which have any one digit that is the average of the other two.

b-

$$b+c=6 \quad | \quad b+c=8$$

$$(113) \quad b+c=10 \quad b+c=12$$

$$b+c=14$$

$$30$$

$$40$$

$$30$$

$$13$$

$$\begin{array}{c} 3 \\ 2 \\ 1 \end{array} \quad \boxed{abc} \quad a = \frac{b+c}{2}$$

or

$$b = \frac{c+a}{2}$$

$$c = \frac{a+b}{2}$$

| Any | Cases                   | Total Cases  |
|-----|-------------------------|--------------|
| 1   | 102, 111                | 4+1=5        |
| 2   | 240, 231, 222           | 4+6+1=11     |
| 3   | 360, 351, 342, 333      | 4+6+6+1=17   |
| 4   | 480, 471, 462, 453, 444 | 4+6+6+6+1=23 |
| 5   | 519, 528, 537, 546, 555 | 6+6+6+6+1=25 |
| 6   | 633, 624, 615, 606      | 6+6+6+1=19   |
| 7   | 795, 786, 777           | 6+6+1=13     |

**Q3.** Find the number of three -digit numbers from 100 to 999 which have any one digit that is the average of the other two.

**Answer: 121**

| Any      | Cases    | Total cases |
|----------|----------|-------------|
| 8        | 897, 888 | $6 + 1 = 7$ |
| 9        | 999      | 1           |
| } 8      |          |             |
| Min. 100 |          |             |

$$b + c = 16$$

$$b + c = 18$$

$$113 + 8$$

$$= 121$$

Solution:

| One of the digits | Pattern                 | Number of numbers        |
|-------------------|-------------------------|--------------------------|
| 1                 | 102, 111                | $4 + 1 = 5$              |
| 2                 | 240, 231, 222           | $4 + 6 + 1 = 11$         |
| 3                 | 306, 315, 324, 333      | $4 + 6 + 6 + 1 = 17$     |
| 4                 | 408, 417, 426, 435, 444 | $4 + 6 + 6 + 6 + 1 = 23$ |
| 5                 | 519, 528, 537, 546, 555 | $6 + 6 + 6 + 6 + 1 = 25$ |
| 6                 | 639, 648, 657, 666      | $6 + 6 + 6 + 1 = 19$     |
| 7                 | 759, 768, 777           | $6 + 6 + 1 = 13$         |
| 8                 | 879, 888                | $6 + 1 = 7$              |
| 9                 | 999                     | 1                        |
|                   | Total                   | 121                      |

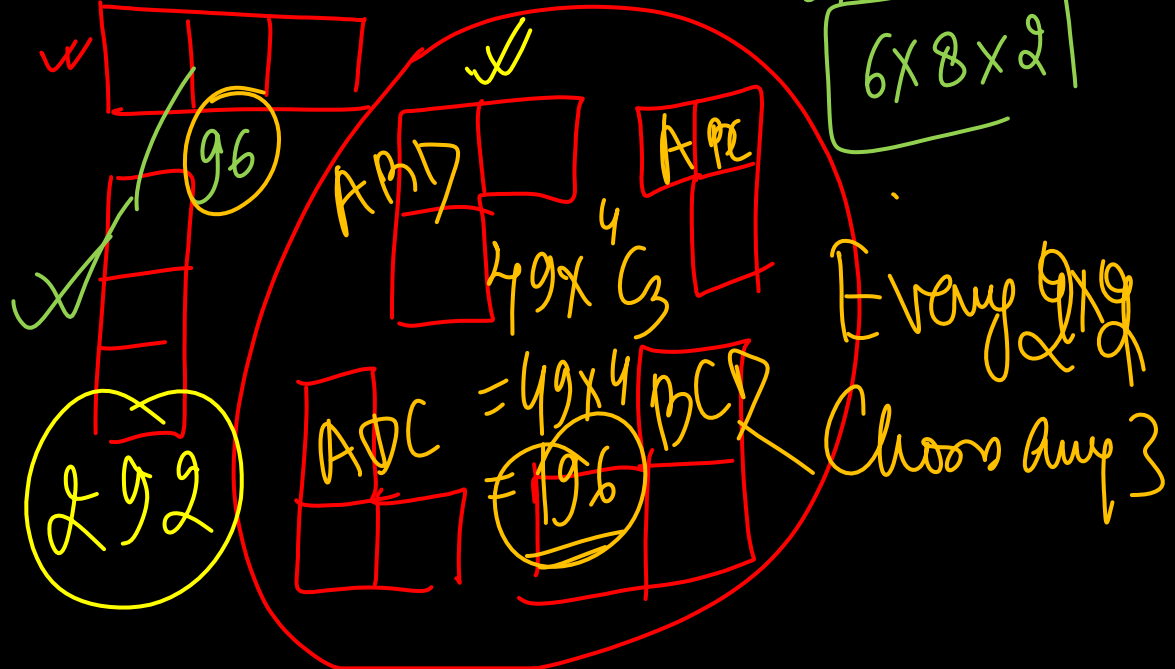
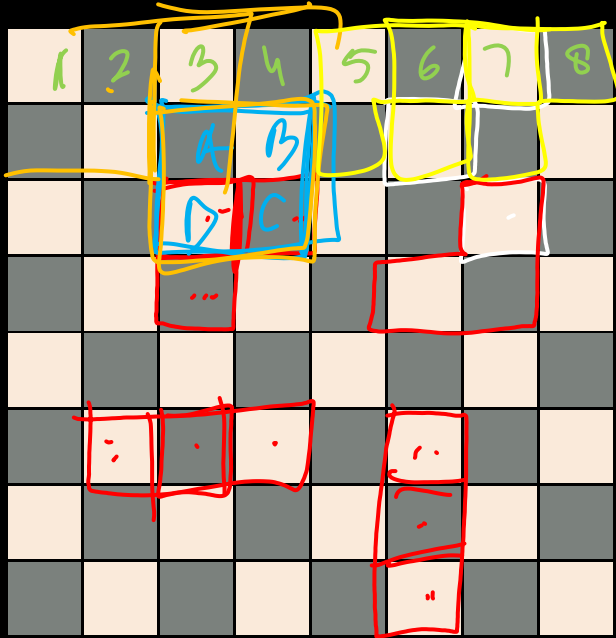
$$1 \times 1 - 8^2 = 64 \quad | \quad 2 \times 2 - 7^2$$

**Q4.** Find the number of ways in which we can choose 3 squares on a chess board such that one of the square has its two sides common to other two square.

123  
234  
345  
456  
567  
678

6

$6 \times 8 \times 2$



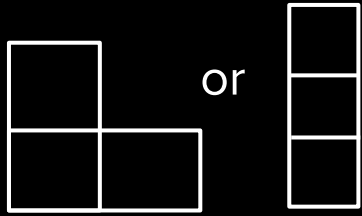
**Q4.** Find the number of ways in which we can choose 3 squares on a chess board such that one of the square has its two sides common to other two square.

**Answer: 292**

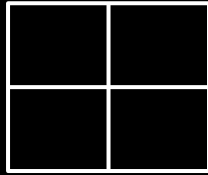


## **Solution:**

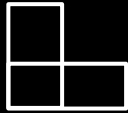
We have to choose following patterns:



Every square of (2 by 2)



Will contribute four



shaped figure by removing any one square out of it. Number of ways to choose these square =  $7 \times 7 = 49$ .



So total

$$\text{Shaped figure} = 49 \times 4 = 196.$$

In every line it is possible to have 6



shaped figure

So total number of such figure  
 $= 6 \times 8 \times 2 = 96$ .

( $\because$  there are eight horizontal lines and eight vertical lines)

$$\therefore \text{Total number of cases} \\ = 196 + 96 = 292$$

$E-M$

$$a=1, 10+b$$

PRMO 2019

2020

Q5. Let  $N$  be the number of ways of choosing a subset of 5 distinct numbers from the set  $\{10a + b; 1 \leq a \leq 5, 1 \leq b \leq 5\}$  where  $a, b$  are integers, such that no two of the selected numbers have the same units digit and no two have the same ten digit. What is the remainder when  $N$  is dividing by 73?

25

$$a=2$$

$$20+b$$

$$\begin{array}{r} 20 \\ 73 \overline{) 20} \\ \underline{73} \\ 47 \end{array}$$

|    |    |    |    |    |
|----|----|----|----|----|
| 11 | 12 | 13 | 14 | 15 |
| 21 | 22 | 23 | 24 | 25 |
| 31 | 32 | 33 | 34 | 35 |
| 41 | 42 | 43 | 44 | 45 |
| 51 | 52 | 53 | 54 | 55 |

very no diff row  
diff column

$$N = {}^5C_1 \cdot {}^4C_1 \cdot {}^3C_1 \cdot {}^2C_1 \cdot {}^1C_1 = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$$

**Q5.** Let  $N$  be the number of ways of choosing a subset of 5 distinct numbers from the set  $\{10a + b; 1 \leq a \leq 5, 1 \leq b \leq 5\}$  where  $a, b$  are integers, such that no two of the selected numbers have the same units digit and no two have the same ten digit. What is the remainder when  $N$  is dividing by 73?

**Answer: 47**

### **Solution:**

$$10 \mid a + b; \quad 1 \leq a \leq 5, \quad 1 \leq b \leq 5$$

Let us divided numbers into different sets, such as

$$\text{Set 1} = \{11, 12, 13, 14, 15\}$$

$$\text{Set 2} = \{21, 22, 23, 24, 25\}$$

$$\text{Set 3} = \{31, 32, 33, 34, 35\}$$

$$\text{Set 4} = \{41, 42, 43, 44, 45\}$$

$$\text{Set 5} = \{51, 52, 53, 54, 55\}$$

Now to make a number having no two digits and no two ten's digit same, we can select any 1 number form each of set 1, set 2, set 3, set 4, set 5.

$$\therefore \text{No of ways} = {}^5C_1 \times {}^4C_1 \times {}^3C_1 \times {}^2C_1 \times {}^1C_1 = 120$$

$$\therefore 120 \div 73$$

$$\Rightarrow \text{Remainder} = 47$$

**Q6.** Three couples sit for a photograph in 2 rows of three people each such that no couple is sitting in the same row next to each other or in the same column one behind the other. How many arrangements are possible?

A 48

B 56

C 72

D 96

$\begin{pmatrix} a & A \\ b & B \\ c & C \end{pmatrix}$

$\boxed{6!}$

no condition

$\begin{pmatrix} c & b & a \\ a & b & A \end{pmatrix}$

$\boxed{\text{Same row}}$

$$D_n = n! \left[ 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^{n-1} \frac{1}{n!} \right]$$

$\begin{pmatrix} A & B & C \end{pmatrix}$

$\boxed{D_2 = 2}$

$\begin{pmatrix} B & C & A \\ C & A & B \end{pmatrix}$

$\boxed{\begin{pmatrix} a & b & c \end{pmatrix}}$

$$2 \cdot 2 \cdot 2 \cdot 3! \cdot 2$$

$$8 \times 6 \times 2 = 96$$

**Q6.** Three couples sit for a photograph in 2 rows of three people each such that no couple is sitting in the same row next to each other or in the same column one behind the other. How many arrangements are possible?

**A** 48

**B** 56

**C** 72

**D** 96

### Solution:

**Case 1:** If one person from each couple placed in 1<sup>st</sup> row.

Number of ways to place in first row =  $6 \times 4 \times 2 = 48$

Correspondingly 2<sup>nd</sup> row fill in 2 ways.

Number of ways =  $48 \times 2 = 96$

|       | $C_1$ | $C_2$ | $C_3$ |
|-------|-------|-------|-------|
| $R_1$ | _____ | _____ | _____ |
| $R_2$ | _____ | _____ | _____ |

Case 2: If one couple placed in 1st row. No such case possible

Total = 96.

**Q7.** Let  $a, b, c$  be distinct positive integers such that  $b + c - a, c + a - b$  and  $a + b - c$  are all perfect squares. What is the largest possible value of  $a + b + c$  smaller than 100?

91 Amazing Q  
81, 119 — 91  
✓

$$\begin{aligned} b + c - a &= x_1^2 \\ c + a - b &= x_2^2 \\ a + b - c &= x_3^2 \end{aligned}$$

$$a + b + c = x_1^2 + x_2^2 + x_3^2$$

$$\{1, 9, 25, 49, 81\}$$

$$(a + b + c)_{\max} < 100$$

$$1, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2, 8^2, 9^2$$

$$a + b - c, b + c - a, c + a - b$$

$$\{4, 16, 36, 64\} \text{ Same parity}$$

$$4 + 16 + 64 = 84$$



**Q7.** Let  $a, b, c$  be distinct positive integers such that  $b + c - a, c + a - b$  and  $a + b - c$  are all perfect squares. What is the largest possible value of  $a + b + c$  smaller than 100?

**Answer: 91**

## **Solution:**

$b + c - a, c + a - b, a + b - c$ ... integers

Add these three

$$b + c - a + c + a - b + a + b - c = a + b + c$$

$$\therefore a + b + c < 100$$

$\therefore$  possible value of  $b + c - a, c + a - b, a + b - c$   
are 1, 4, 9, 16, 25, 36, 49, 64, 81.

Either  $b + c - a, a + b - c, c + a - b$  all three will be odd or all will be even.

$\therefore$  Largest possible value of  $a + b + c$  is possible.

If

$$b + c - a = 81$$

$$c + a - b = 9$$

$$a + b - c = 1$$

$$\therefore a + b + c = 91$$

Q8. What is the number of triples  $(a, b, c)$  of positive integers such that

(i)  $a < b < c < 10$  and

(ii)  $a, b, c, 10$  form the sides of a quadrilateral?

$38 + 25 + 10 = 73$

$a + b + c > 10$

| a | b | c          |
|---|---|------------|
| 1 | 2 | 8, 9       |
|   | 3 | 7, 8, 9    |
|   | 4 | 6, 7, 8, 9 |
|   | 5 | 6, 7, 8, 9 |
|   | 6 | 7, 8, 9    |
|   | 7 | 8, 9       |

| a | b | c             |
|---|---|---------------|
| 1 | 8 | 9             |
| 2 | 3 | 6, 7, 8, 9    |
| 2 | 4 | 5, 6, 7, 8, 9 |
| 2 | 5 | 6, 7, 8, 9    |
|   | 6 | 7, 8, 9       |
|   | 7 | 8, 9          |
|   | 8 | 9             |

| a | b | c             |
|---|---|---------------|
| 3 | 4 | 5, 6, 7, 8, 9 |
|   | 5 | 6, 7, 8, 9    |
|   | 6 | 7, 8, 9       |
|   | 7 | 8, 9          |
|   | 8 | 9             |
| 4 | 5 | 6, 7, 8, 9    |
|   | 6 | 7, 8, 9       |
|   | 7 | 8, 9          |

| a | b | c       |
|---|---|---------|
| 5 | 6 | 7, 8, 9 |
|   | 7 | 8, 9    |
|   | 8 | 9       |
| 6 | 7 | 8, 9    |
|   | 8 | 9       |
| 7 | 8 | 9       |

Systematic  
Counting

Q.

AATE HAIN!

**Q8.** What is the number of triples  $(a, b, c)$  of positive integers such that

(i)  $a < b < c < 10$  and

(ii)  $a, b, c, 10$  form the sides of a quadrilateral?

**Answer: 73**

**Solution:**

$a + b + c > 10$   
 $\therefore (a, b, c)$  can be

| a | b | c       |
|---|---|---------|
| 1 | 2 | 8,9     |
| 1 | 3 | 7,8,9   |
| 1 | 4 | 6,7,8,9 |
| 1 | 5 | 6,7,8,9 |
| 1 | 6 | 7,8,9   |
| 1 | 7 | 8,9     |
| 1 | 8 | 9       |
| 2 | 3 | 6,7,8,9 |

| a | b | c         |
|---|---|-----------|
| 2 | 4 | 5,6,7,8,9 |
| 2 | 5 | 6,7,8,9   |
| 2 | 6 | 7,8,9     |
| 2 | 7 | 8,9       |
| 2 | 8 | 9         |
| 3 | 4 | 5,6,7,8,9 |
| 3 | 5 | 6,7,8,9   |
| 3 | 6 | 7,8,9     |
| 3 | 7 | 8,9       |
| 3 | 8 | 9         |
| 4 | 5 | 6,7,8,9   |

| a | b | c     |
|---|---|-------|
| 4 | 6 | 7,8,9 |
| 4 | 7 | 8,9   |
| 4 | 8 | 9     |
| 5 | 6 | 7,8,9 |
| 5 | 7 | 8,9   |
| 5 | 8 | 9     |
| 6 | 7 | 8,9   |
| 6 | 8 | 9     |
| 7 | 8 | 9     |

Total cases = 73



Homework  
**SAWAAL**

1 2 3 X  
1 2 4 X

14

- |   |      |
|---|------|
| A | 2148 |
| B | 2268 |
| C | 2358 |
| D | 2468 |

**Note:** Solution at last



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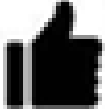
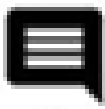
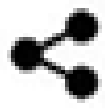
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**GOLD**  
For  
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**Q9.** Suppose 28 objects are placed along a circle at equal distances. In how many ways can 3 objects be chosen from among them so that no two of the three chosen objects are adjacent nor diametrically opposite?

**A** 2148

**B** 2268

**C** 2358

**D** 2468

### **Solution:**

One can choose 3 objects out of 28 objects in  $\binom{28}{3}$  ways. Among these choices all would be together in 28 cases: exactly two will be together in  $28 \times 24$  cases. Thus three objects can be chosen such that no two adjacent in  $\binom{28}{3} - 28 - (28 \times 24)$  ways. Among these, further, two objects will be diametrically opposite in 14 ways and the third would be on either semicircle in a non adjacent portion in

$28 - 6 = 22$  ways. Thus required number is

$$\binom{28}{3} - 28 - (28 \times 24) - (14 \times 22) = 2268.$$